Study Proposal for KTU Identification *

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Executive Summary

This report evaluates a wound-charting dataset to determine whether it can support a statistical test of the Kennedy Terminal Ulcer (KTU) hypothesis and, having found it cannot, to specify a future study design that can. The key finding is that the current data are not analyzable for KTUs because discharge outcomes are frequently missing or ambiguous, and many documented discharges are not linked to a verified death time, making end of life comparisons biased and unreliable. Exploratory visualizations still suggest potentially meaningful structure, including differences between sacral and non-sacral wound timelines and stronger end of life indicators among patients with sacral wounds, but these trends cannot be interpreted definitively under the present missingness and selection concerns. To resolve this, the report recommends a prospective nursing home study designed to test if wounds occur at higher rates approaching end of life after adjusting for confounding factors. Using data from the 1989 Byron Health Center report, a one year target of roughly 3,000 beds was determined for the study, providing sufficient end of life event data for the models. If terminal acceleration is confirmed, the study will then use clustering on standardized wound covariates to derive an empirical definition of KTU-like wounds, avoiding the need for currently uncertain medical diagnosis while directly informing the client's concerns about caretaker liability. Supplementary diagnostics and figures are provided in the Appendix.

(1) Introduction

The goal of this report is to evaluate the provided dataset to design a future study capable of overcoming current data limitations. This proposed study is designed to be entirely driven by statistical methodology rather than traditional medical diagnosis. Since the Kennedy Terminal Ulcer (KTU) is a theorized phenomenon that cannot yet be confirmed through standard medical testing, we must use statistical modeling to validate its existence as an abnormal occurrence.

The implications of this report are significant, particularly regarding the second motivating question: "Can uniform care prevent these pressure-like wounds from occurring?". This analysis investigates the extent to which providers can be held responsible for wound development. This carries substantial real-world stakes, as caregivers have faced criminal charges for wounds sustained by patients; wounds that, under theoretical standard circumstances, they were expected to prevent.

(2) Data

The main variables of interest in this analysis are `Date of Service` (when a wound was treated), `MRN` (unique patient identifier), `Admission Date`, `Discharge Date`, and `Discharge Reason`. From these variables, we can identify the frequency of wounds in relation to how close a patient is to end of life or other discharge reasons. While this is the core concept of the analysis, additional covariates like `Wound Status`, `Wound Measurements`, and `Wound Anatomical Site` will also be useful for clustering the wounds into distinct groups and validating the argument for uniformity of care.

*Formatting based on svmille guide (http://github.com/svmiller).

One of the limitations of the data provided for this analysis is that entries for three of the main variables referenced are often missing, as seen in Figure 1.

Missing Data Concerns

Since the purpose of analyzing these records would be to validate whether KTUs exist and to characterize their features, the discharge process must be unambiguous. In the current dataset, however, discharge is not consistently tied to end of life outcomes, and in many cases the patient is effectively lost to follow-up. Nearly half of patients have missing values for `Discharge Date` and `Discharge Reason`, and this missingness is not interpretable: we cannot determine what fraction is missing because the patient was still under the facility's care (and therefore no discharge entry was made), what fraction reflects unrecorded departures as noted in the meeting with the clients, or what fraction is due to other uncontrolled causes. Even among patients with documented discharge information, 34 were not deceased but instead were discharged to hospice, transferred to another facility or provider, or left for other reasons. Because the KTU hypothesis is explicitly about wounds that emerge shortly before end of life, these incomplete outcomes make inference impossible. For non-deceased discharges and for those lost to follow-up, we do not know if or when death occurred after leaving observation. As a result, any conclusion drawn just from the subset of deceased patients would be biased, because it conditions on an outcome that is unknown for a substantial portion of the records and therefore cannot be compared in a valid, representative way.

Data Collection Concerns

A second limitation is the manner in which wound information was collected and categorized. The variable `Wound Acquired At` contains 63 different values, including broad categories such as "unknown" and situations that are plausibly unrelated to pressure wound (ie: "home after a fall"), which likely reflect different wounds than the typical bed sores that KTUs are meant to be distinguished from. Without a controlled setting and standardized documentation of when each wound occurred and why it occurred, it is not possible to separate pressure wounds from unrelated wounds. A supervised environment such as a nursing home would allow consistent charting of all wounds whenever a patient is turned over, producing a representative sample. In contrast, if the current dataset was assembled largely through referrals triggered by concerns that KTU-like wounds were present, the sample may overrepresent suspicious cases and underrepresent end of life patients who never develop a KTU-like wound. That kind of selection process would bias any estimated frequency or distinguishing characteristics, further preventing a proper analysis.

(3) Data Visualization

Regardless of the previously discussed data limitations, specific visualizations can be used to inform and motivate the design of a future study. These visuals will not definitely prove the existence of KTUs, but the trends do give some evidence.

Wound Location

Because KTUs present in the sacral region more often than other regions, it is important to analyze whether wounds in this area exhibit distinct characteristics or outcomes compared to non-sacral wounds. Initial evidence from Figure 2 reveals that the notched confidence intervals for the medians do not overlap. This provides solid statistical evidence for the idea that sacral and non-sacral wounds operate on different timelines, suggesting that some sacral wounds may be unique from a standard pressure wound. Supporting this, patients with sacral wounds are approximately three years older on average and are four times more likely

to have hospice documentation, which are both strong indicators of proximity to end of life. However, Figure 3 highlights the persistent impact of data missingness: while sacral wounds are more prevalent among patients with documented deceased or discharged outcomes, nearly half of the total dataset consists of patients with unknown statuses, where non-sacral wounds are slightly more common. Additionally, while the sacrum is one of the most common locations for a KTU, this does not imply that all sacral wounds are KTUs. Therefore, while wound location remains a promising identifier for KTUs, without more clarity about patient outcomes we cannot give any definitive conclusions.

Patient Age and Wound Area

Further analysis explores the relationship between a patient's age at the time of wound and the measured wound area (Figure 4). The resulting density plot does not reveal a clear linear correlation between age and wound size; instead, the observations within the various density contours contain relatively equal proportions of both end of life patients (those deceased or in hospice) and those without such documentation. A notable cluster of end of life patients exists around age 89, specifically within the highest density "yellow" contour, which may point toward the presence of terminal wounds appearing shortly before death. Nevertheless, this remains relatively weak evidence of a specific KTU characteristic. Also, since we cannot verify the actual survival duration of patients after entering hospice, it is impossible to confirm if these wounds align with the rapid, terminal trajectory characteristic of a KTU.

(4) Study Proposal

Given the limitations of the provided dataset, this section outlines a new study design that would avoid those issues and allow us to conclusively answer two motivating questions:

(4.1) Do patients develop pressure-like wounds at an increased rate near end of life?

If the phenomenon reported in Karen Kennedy's 1989 report (namely that some patients develop pressure-like wounds shortly before death) truly exists, it can be tested without relying on medical assumptions.[1] The key requirement is to show that wound occurrence is not consistent over time within the same individuals, but instead increases disproportionately as end of life approaches.

(4.2) Can uniform care prevent these pressure-like wounds from occurring?

A second objective is to evaluate Michael Miller's argument that, even if end of life physiology (like skin failure) increases vulnerability, it does not explain why wounds would concentrate in specific anatomical regions, since systemic decline should affect all body tissue.[2] Under this view, the most plausible explanation for region-specific wounds is still localized pressure; this implies that uniform care could prevent these wounds in the same way it prevents typical pressure wounds, and therefore that providers may remain liable if such wounds occur.

Study Design

The motivating context for this study design is the data from the 1989 report from Byron Health Center (a 500 bed intermediate care facility in Fort Wayne). The proposed study would be conducted over the course of a single year in a few nursing homes such that a target number of $3,000$ beds in the study could be attained. The reasoning behind this target using data from Table I in the 1989 report is as follows:

- Total deaths across the 5 years of the 1989 report: $83 + 89 + 111 + 85 + 101 = 469$
- Deaths with pressure sores across those 5 years: $3 + 2 + 10 + 15 + 21 = 51$

[1] Karen (1989) book.
[2] Michael (2017) book.

So the conditional proportion of deaths that had pressure sores is:

$$P(\text{pressure sore} \mid \text{death}) = \frac{51}{469} \approx 10.9\%$$

- Average deaths per year is: $\frac{469}{5} = 93.8$
- Annual death fraction per occupied bed is: $\frac{93.8}{500} \approx 18.8\%$
- Probability that a bed yields a death with pressure sores is: $0.1876 \times 0.1087 \approx 2.0\%$

Additionally, based on the excerpt "The 51 persons who died . . . had a total of 95 pressure wounds," then:

- Pressure wounds per pressure sore death is: $\frac{95}{51} = 1.86$

A conservative rule for the survival model used in the next section is approximately 50 to 60 deaths with pressure sores (this is the event of interest). If we observe N beds for a year, the expected number of deaths with pressure sores would be:

$$E[M] \approx 0.0204 \times N$$

Setting $E[M] = 60$ gives:

$$N \approx \frac{60}{0.0204} \approx 2941$$

beds

Rounding up gives $N \approx 3,000$, which implies the following for the rest of the data:

- Expected total deaths in one year: $E[D] \approx 0.1876 \times 3000 \approx 563$
- Expected deaths with pressure sores: $E[M] \approx 0.0204 \times 3000 \approx 61$ (consistent with target)
- Expected number of pressure wounds among those pressure sore deaths: $E[\text{wound events in deceased}] \approx 1.86 \times 61 \approx 113$

Given the expectation of over 100 wound events in the deceased, this will likely be sufficient for modeling differing rate of wound events leading up to end of life. A target of $3,000$ beds is equivalent to roughly $\frac{3,000}{500} = 6$ facilities of similar size to the Fort Wayne facility. It is important to remember that this study could also be extended to two years and have a target number of $1,500$ beds and so on.

The data required for this study is the same as those outlined in the data section: `Date of Service`, `MRN`, `Admission Date`, `Discharge Date`, and `Discharge Reason`. Similarly to the provided dataset, additional covariates about the wound itself will be used for clustering the observed wounds after the first two motivating questions are addressed. In regards to the second motivating question about uniform care, a more standardized wound charting system would be as follows:

- Patient ID (`MRN`)
- Wound discovery time (`Date of Service`)
- Turning position of the patient (ie: supine, prone, etc) taken from a separate log that is updated every time a patient is turned

- Any additional wound observations already being made, such as Wound Measurements

The only additional information that is being included here is the turning position and corresponding position logs, which are necessary to disproving patient neglect brought up in the second motivating question. Lastly, data concerning the general health of residents like mobility and age will be needed for one of the models. The necessity of strict study control cannot be overstated. Within the cohesive, closely monitored environment of a nursing home, we can move beyond a general focus on residents who exhibit suspected patterns. This controlled setting allows for a fully representative sample, ensuring that all residents are recorded and all wounds are documented using a standardized methodology. This consistency is essential to eliminate the biases that currently prevent our ability to verify the theorized phenomenon.

(4.1) Question One

To determine whether residents develop pressure-like wounds at an increased rate near the end of life, we need a model that can estimate how the instantaneous rate of new wound occurrence changes as a resident approaches death. A standard Poisson model is not sufficient because it effectively treats wound events as occurring at a roughly stable rate over time, conditional on measured risk factors. That approach breaks down here for two reasons.

First, the hypothesis we want to test is explicitly about a terminal acceleration, and the last days or weeks of life may be physiologically different from the rest of a resident's stay. A standard Poisson model can only incorporate closeness to death if death time is known, which is not possible at the time wounds occur and is not available for residents who do not die during observation. Second, a typical nursing home study contains substantial right-censoring: many residents are still alive when the study ends or are discharged. For those residents, the death time T_i is not observed, so we cannot directly compute time to death during their observed follow-up. Ignoring these residents would waste most of the data and can bias the results because the people who die during follow-up are systematically different from those who do not.

One way to handle both issues is to fit a joint model that simultaneously describes the resident's risk of death and the timing of wound occurrences, with the two processes linked through a shared resident-specific term called frailty (u_i). Frailty captures unmeasured vulnerability that affects both outcomes; without it, the model could incorrectly attribute the effect of frail residents having more wounds to residents close to death have more wounds, even when those are partially distinct effects.

End of Life Model

The end of life component models the resident's hazard of death: the instantaneous risk that death occurs at time t, given survival up to time t:

$$h_i(t) = h_0(t) \exp(\gamma^\top X_i(t) + \alpha u_i)$$

Here, $h_0(t)$ is the baseline hazard (death risk over time for a reference resident), $X_i(t)$ are measured covariates (ie: mobility status and age), and u_i is the resident-specific frailty. The coefficient α quantifies how strongly frailty influences mortality: if $\alpha > 0$, higher frailty increases death risk. This submodel is similar to a Cox Proportional Hazards model, and can be fit by maximizing the likelihood that the observed sequence of deaths happened exactly as it did. This leads to parameter estimates for γ and α, remembering to treat residents who are alive at study end as right-censored (and similarly treating discharges as censored unless mortality is known).

Wound Model

The wound component models the rate at which new wounds occur at time t.

$$\lambda_i(t) = \lambda_0(t)\exp(\beta^\top X_i(t) + u_i + f(\tau_i(t))), \quad \tau_i(t) = T_i - t$$

$\lambda_0(t)$ is the baseline wound occurrence rate, $\beta^\top X_i(t)$ adjusts for measured risk factors, and the same frailty u_i appears again to allow some residents to be persistently more wound-prone than others. The key term is $f(\tau)$, where $\tau_i(t) = T_i - t$ is time to death for residents whose death time is observed. This function $f(\tau)$ is designed to capture a spike in wound risk as death approaches: when τ is large (death far away), we expect $f(\tau)$ to be near zero; when τ is small (death imminent), $f(\tau)$ may be positive and large.

Because we do not want to assume an arbitrary shape for the terminal effect, we represent $f(\tau)$ flexibly as a spline (a weighted sum of smooth basis functions):

$$f(\tau) = \sum_{m=1}^{M} \eta_m B_m(\tau)$$

and estimate the coefficients $\eta_1, \ldots, \eta_M$ from the data. Practically, this lets the data determine whether the curve is flat, gradually increasing, or sharply rising as $\tau \to 0$. To ensure interpretability and avoid overfitting, we pre-specify a reasonable knot placement (ie: more flexibility in the last 0 to 14 days and less beyond 60 days) and apply standard smoothing as needed. Just like the other model, we can fit this model by finding the parameters (β, η, α) that achieve the maximum likelihood. Intuitively, we can imagine the model evaluating every recorded wound to determine its likely cause: the resident's general frailty (u_i), a specific risk factor like low mobility (X_i), or proximity to the end of life ($f(\tau)$). The inclusion of frailty (u_i) in both submodels is critical for balancing these signals. For example, if a resident develops five wounds but survives for three years, the Wound Model identifies them as "wound-prone," while the End of Life model confirms they are not "death-prone". By reconciling these two signals, the model can effectively isolate the true terminal spike from general vulnerability and determine whether an additional terminal effect $f(\tau)$ is still necessary to explain the timing pattern of wound occurrence near death.

For residents who die during the study, T_i is observed, so $\tau_i(t)$ is known and directly informs $f(\tau)$. For residents who do not die during observation, T_i is unknown, so $\tau_i(t)$ cannot be computed directly. The joint model resolves this by using the End of Life model to represent uncertainty about T_i. Given the resident's survival until the censoring date, their specific risk factors, and their calculated frailty, the model assumes their most likely death time occurs after the study ended. In estimation, the wound likelihood contribution for censored residents is computed by integrating over plausible death times consistent with their observed survival. This allows censored residents to contribute information about baseline and covariate-driven wound rates without falsely treating them as if they were in a terminal period.

Hypothesis Test

The primary goal is to assess whether the rate of wound occurrences accelerates as death approaches, which corresponds to the shape of $f(\tau)$.

- Null Hypothesis: $f(\tau) = 0$ for all τ (wound risk depends on baseline rate, covariates, and frailty, but not on proximity to death)
- Alternative Hypothesis: $f(\tau)$ increases as $\tau \to 0$ (after accounting for covariates and frailty, wound risk still rises as death approaches)

While the clearest presentation is a visual plot of $f(\tau)$ showing the terminal multiplier over time, we can formally test this hypothesis using a likelihood ratio test. This test compares the full joint model (which includes the spline coefficients $\eta_1, \ldots, \eta_M$) against a reduced model where $f(\tau)$ is forced to zero. If the full model fits the data significantly better, we have strong statistical evidence that patients do develop pressure-like wounds at an increased rate near end of life.

(4.2) Question Two

This second section assumes that a confirming result was found for the first motivating question. To determine whether uniform care can prevent these pressure-like wounds from occurring, we must address the challenge of uniform care. Michael Miller argues that wounds appearing near the end of life are simply ordinary pressure wounds resulting from inadequate repositioning. To refute this, the analysis must examine wound occurrence after each scheduled turn (typically every 2 to 3 hours). The goal is to demonstrate that KTU-like wounds still appear even when turning protocols are followed and the anatomical site is properly offloaded.

Interval Hazard Model

To analyze these scheduled turns, we convert each resident's timeline into a series of turning intervals $[t_{ik}, t_{i(k+1)})$. Each interval is assigned specific exposure variables and an outcome indicator. We then model the probability that a new wound first appears in a given interval k using a discrete-time hazard model:

$$\Pr(Y_{ik} = 1) = g^{-1}\Big(\theta_0 + \theta_1 D_{ik} + \theta_2 Z_{ik} + \theta_3 P_{ik} + b_i + b_{\text{home}}\Big)$$

In this model, D_{ik} represents dependent exposure (ie: the number of minutes the eventual wound site was plausibly under pressure), P_{ik} indicates protocol deviations (ie: late or undocumented turns), and Z_{ik} represents terminal proximity (proximity to death). The terms b_i and b_{home} are random effects that account for resident-specific frailty and facility-level differences, respectively.

This structure allows us to isolate the influence of care from the influence of physiology. If $\theta_1 > 0$ and $\theta_3 > 0$, the model confirms that pressure and missed turns increase risk. However, the critical test for this hypothesis is whether $\theta_2 > 0$. A positive θ_2 indicates that, even after controlling for pressure and missed turns, the risk of wound occurrence still rises significantly as a resident approaches death.

Next, we examine rapid progression. Even if the timing of the initial occurrence is debated, the speed at which a wound worsens is highly measurable. We fit a new survival model to the time it takes for a wound to transition from first detection to a severe stage:

$$h(t) = h_0(t) \exp(\psi_1 D_{ik} + \psi_2 Z_{ik} + \psi_3 P_{ik})$$

If $\psi_2 > 0$, it indicates that wounds worsen faster near death even when repositioning is identical to earlier periods of care.

Hypothesis Test

The primary goal of this analysis is to determine if terminal physiology contributes to wound occurrence independently of repositioning. This test evaluates the significance of terminal proximity across both the interval hazard and rapid progression models.

- Null Hypothesis: wound occurrence and progression are explained entirely by pressure exposure (D_{ik}) and protocol deviations (P_{ik}) (terminal proximity (Z_{ik}) has no independent effect, meaning $\theta_2 = 0$ and $\psi_2 = 0$)

- Alternative Hypothesis: terminal proximity (Z_{ik}) significantly increases the risk of wound occurrence and the speed of progression (even when care is uniform and the site is offloaded)

Statistical significance is determined by assessing the coefficients for terminal proximity (θ_2 and ψ_2) using a likelihood ratio test. If these parameters are found to be significantly greater than zero, it provides rigorous evidence that the terminal spike contributes to skin failure beyond the effects of missed turns or physical pressure. Furthermore, observing a non-trivial frequency of wounds appearing on sites that were documented as offloaded during the prior turning logs would further refute the claim that these are standard pressure wounds. Such findings would demonstrate that while uniform care is important for preventing traditional pressure wounds, it cannot fully prevent wounds associated with the end of life process.

Clustering

Once again assuming the first motivating question is supported, we can now determine whether the excess wounds occurring near end of life form a distinct, repeatable phenotype that could plausibly correspond to KTUs. Importantly, this step does not rely on a formal medical diagnosis, of which there is none. Instead, it uses the study's standardized wound-level covariates to identify whether the terminally clustered wounds share common characteristics that distinguish them from typical pressure wounds. The guiding principle is: if a terminal phenomenon exists, then wounds that appear during the period where $\exp(f(\tau))$ is elevated should not be an arbitrary mix of unrelated wounds, but should exhibit consistent characteristics (ie: location patterns, rapid progression, etc) that emerge as a cluster.

First, we use the fitted terminal multiplier $\exp(f(\tau))$ from the first motivating question to define a terminal window in an empirically justified way rather than choosing an arbitrary cutoff. For example, we may define the terminal-risk period as the set of τ values where $\exp(f(\tau))$ exceeds a pre-specified threshold (ie: at least 2 times the baseline) or where the curve shows its steepest increase. Wounds that first appear within this terminal-risk period for the deceased form the target set, while wounds occurring far from death form a comparison set that reflects baseline pressure wound behavior. Second, within the target set, we can use standard algorithms like hierarchical clustering to cluster wounds with variables such as `Wound Anatomical Site`, `Wound Measurements`, `Wound Status`, and other standardized descriptors, while excluding any direct use of death time or hospice status in the clustering inputs. This avoids circularity and makes sure that any identified cluster is driven by wound characteristics rather than by the outcome label. After clustering the data in the comparison set, we can compare. In practice, we should have a small number of candidate wound phenotypes (clusters) that can be summarized by interpretable feature profiles (ie: sacral location, butterfly shape, etc). A cluster that is disproportionately represented when $\exp(f(\tau))$ is high provides an empirical candidate definition for a KTU-like wound.

(5) Conclusions

The main conclusion of this report is that the provided dataset cannot support a rigorous statistical test of the KTU hypothesis, but it can still be used to motivate a future study that can. The primary obstacle is a lack of reliable outcome data: nearly half of the records lack a `Discharge Date` or `Discharge Reason`, even documented discharges are often ambiguous. Since the KTU hypothesis relies on wounds appearing shortly before death, the inability to confirm death times for a large portion of the sample prevents valid statistical inference and introduces significant selection bias.

A second conclusion is that the heterogeneity of wound circumstances in the provided dataset limits interpretability. The variable `Wound Acquired At` includes dozens of categories, making it difficult to ensure the comparison of like wounds. Without a controlled environment and standardized charting that

captures when wounds occur, under what conditions they occur, and how they evolve, the data cannot separate typical pressure wounds from terminal skin changes. While exploratory visualizations suggest meaningful differences between sacral and non-sacral timelines, these trends remain speculative due to outcome missingness and potential bias.

Based on these limitations, the main recommendation is to conduct a new prospective nursing home study in a supervised setting targeting approximately $3,000$ beds where outcomes can be recorded consistently. Based on historical event rates, this scale is expected to yield approximately 563 deaths and 60 deaths with pressure-like sores. This volume provides sufficient power to fit the two proposed models: a joint model to test if wound occurrence accelerates as death approaches, and an interval hazard model to test if wounds occur despite uniform repositioning and offloading.

If the first model confirms a terminal acceleration, the study will use wound-level covariates to derive a statistical description of KTU-like wounds through clustering. By defining an empirically justified terminal-risk period and clustering wounds using standardized descriptors, we can determine if end of life wounds form a coherent biological phenotype. If this phenotype is overrepresented among wounds arising under uniform care or at offloaded sites, it would provide the strongest statistical evidence for a distinct terminal wound process.

Appendix

EDA Figures

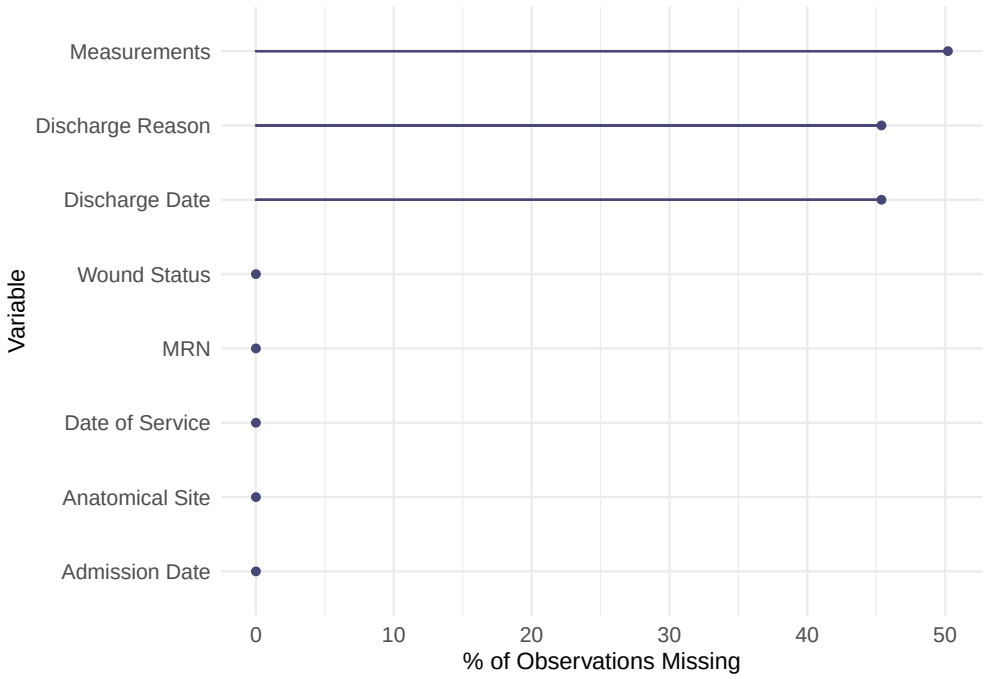

Figure 1: Missing Data for Main Variables

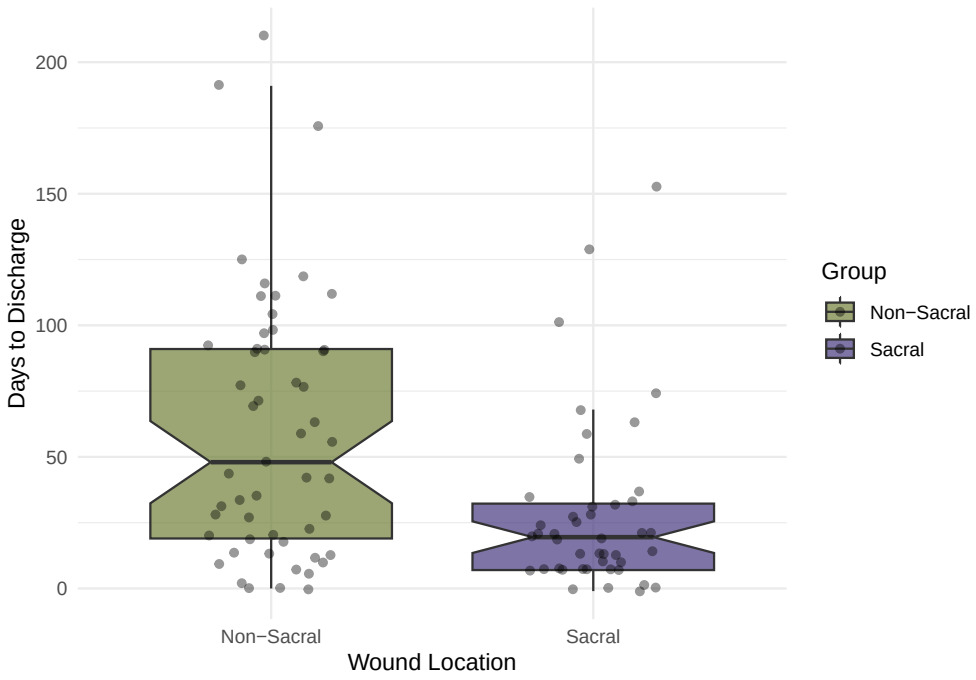

Figure 2: Discharge Time Differences by Wound Location

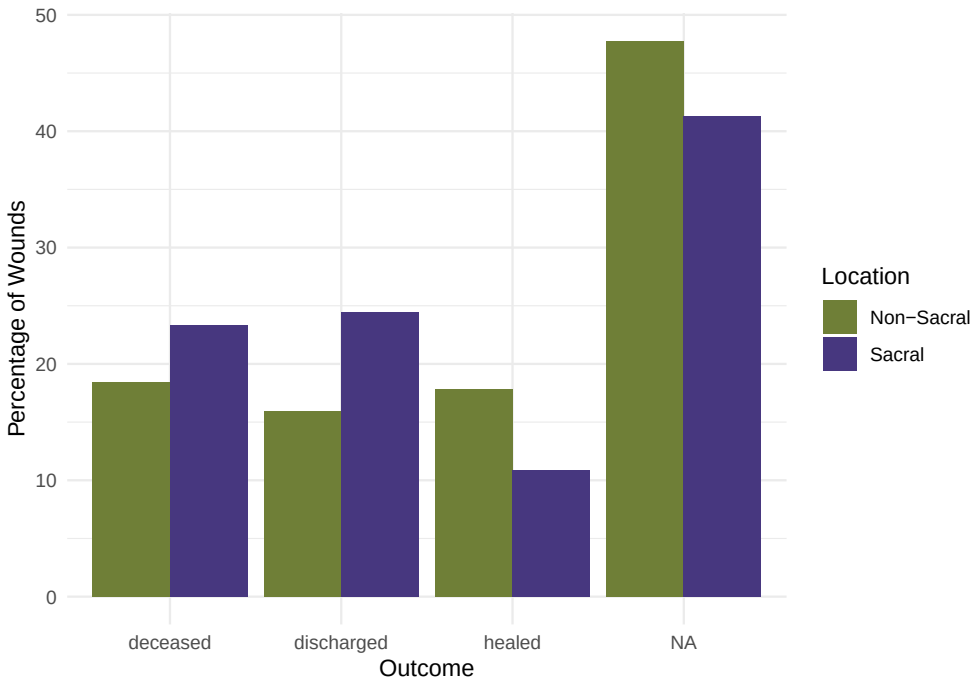

Figure 3: Outcomes by Wound Location

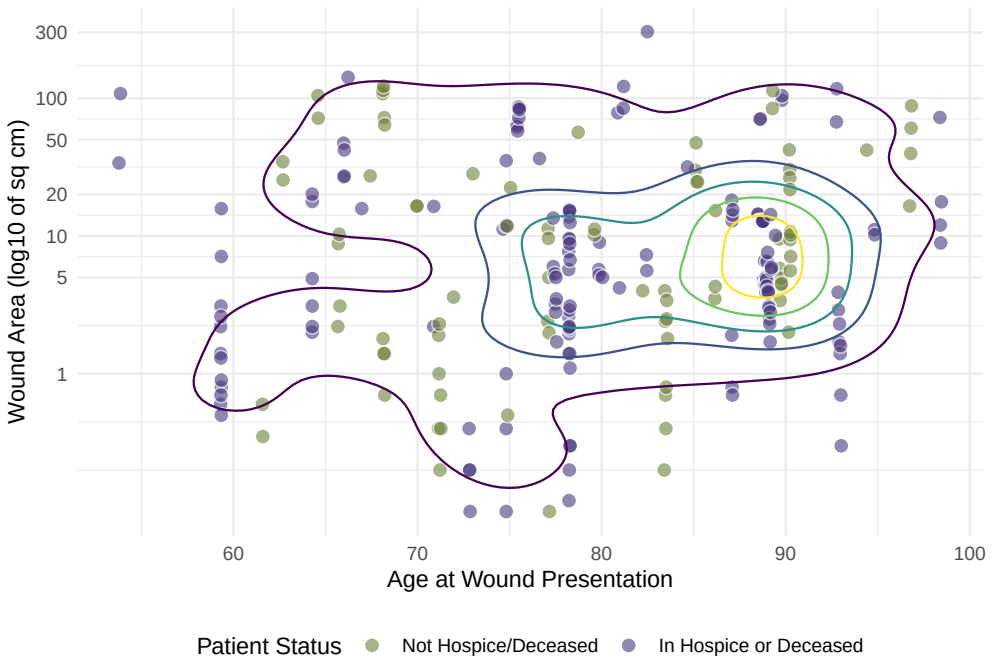

Figure 4: Log Wound Area vs Age

References

Karen, Kennedy. 1989. *THE PREVALENCE OF PRESSURE ULCERS IN AN INTERMEDIATE CARE FACILITY*. DECUBITUS.

Michael, Miller. 2017. *The Death of the Kennedy Terminal Ulcer*. PubMed.